

Pruthvirajsinh Punada

iOS & visionOS Engineer — Apple Platforms

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SUMMARY

Apple-platform software engineer trained at the **Apple Developer Academy** (Naples) and author of **10+ shipped apps** across **iOS, iPadOS, macOS and visionOS** in Swift/SwiftUI. I combine a **Data Analytics** degree with hands-on **on-device ML** (Core ML, Vision) and **3D / spatial computing** (RealityKit, ARKit, Blender→USDZ). Track record includes a Vision Pro chess app integrating the **Stockfish C++ engine** with live online play, and a **production website shipped for a paying client**. Comfortable owning a feature end-to-end and working across the stack.

SKILLS

Languages: Swift, Python, TypeScript / JavaScript, C++, SQL

Apple frameworks: SwiftUI, UIKit, AppKit, RealityKit, ARKit, Core ML, Vision, Combine, SpriteKit, AVFoundation

Platforms: iOS, iPadOS, macOS, visionOS

Machine Learning / Data: Core ML, Vision, PyTorch, TensorFlow, scikit-learn, computer vision, anomaly detection

Web & 3D: React, Next.js, Three.js / React Three Fiber, WebGL, Blender, USDZ / glTF

Tooling: Git, Xcode, Firebase, Vercel, REST APIs, CI/CD, MVVM

SELECTED PROJECTS

LiveChess — Spatial Chess for Apple Vision Pro

Swift · visionOS · RealityKit · C++

Team project (contributions merged into the main repo). Integrated the **Stockfish C++ engine** into Swift for an adjustable-strength AI, move analysis and opening book; built **online play over the Lichess API** (OAuth PKCE, live NDJSON game/event streaming, matchmaking); immersive RealityKit board with custom USDZ pieces, game review and puzzle modes.

PinballVision — visionOS Pinball Game

visionOS · RealityKit · ARKit

Fully playable Vision Pro pinball with **ARKit hand tracking** (pinch to flip), a procedural physics table built from RealityKit bodies, real-time **synthesized audio** (AVAudioEngine), a combo multiplier up to **10x**, and Game Center leaderboards.

SignBridge — On-Device ASL Tutor

Swift · Core ML · Vision

Teaches the full **26-letter ASL alphabet** via **real-time hand-sign recognition at 60 FPS** — Vision hand-pose detection feeding a custom Core ML classifier with prediction smoothing; spell-to-play SpriteKit mini-games. Fully on-device, no servers.

CellularModem-ML — Real-Time Network Anomaly Detection

Python · PyTorch · TensorFlow

Streaming anomaly-detection system across **4 protocol types** (TCP/UDP/IP/QUIC); an ensemble of three models — Isolation Forest, **LSTM (PyTorch)** and **Autoencoder (TensorFlow)** — with **sub-100 ms** inference and a Streamlit monitoring dashboard.

Pizz'Art Caserta — Production Restaurant Website (client)

Next.js · TypeScript · React

Full-stack Next.js site (~98% TypeScript) **shipped for a real, paying restaurant client** and live in production: an online reservation flow handling **~300 reservations/month**, wired to an email pipeline, internationalized + SEO-optimized, deployed on Vercel behind Cloudflare.

BetterFinder — macOS Finder Alternative

Swift · SwiftUI · AppKit

Power-user file manager with a built-in terminal, fast search and syntax-highlighted preview. **Authored the Smart / Batch Rename module** — rule-based renaming with live preview and conflict detection.

EDUCATION

Apple Developer Academy — Università di Napoli Federico II

2025 – 2026 · Naples

iOS Software Engineering & App Foundation Program

University of Campania "Luigi Vanvitelli" — BSc, Data Analytics

2022 – 2026

Thesis: SVM land-cover classification of **27,000 EuroSAT images** into 10 classes (HOG features, comparison of 3 kernels).

HIGHLIGHTS

- **Apple Swift Student Challenge 2026** — submitted *EchoPaint*, an accessibility-first drawing app for blind users (audio + touch).
- **InShorts Italia** — native iOS news aggregator, live on the App Store with **400+ downloads**; aggregates 70+ Italian sources across 9 categories via a Firebase Cloud Functions backend (RSS + AI summarization).
- Spatial-computing focus — multiple shipped **visionOS** apps, a niche with very few engineers at entry level.

LANGUAGES

English (professional) · Italian (fluent)